Architectural Modifications to XFeat for Lightweight Image Matching

Assignment Report — Deep Learning / Computer Vision | Group 48

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1. Base Paper Overview

1.1 Problem and Motivation

Local image feature extraction is central to tasks such as visual localization, camera pose estimation, and 3D reconstruction. State-of-the-art methods like LoFTR, DKM, and SuperGlue achieve high accuracy but at prohibitive computational cost, requiring powerful GPUs and making them unsuitable for edge devices, mobile robots, and augmented reality applications.

1.2 Proposed Solution — XFeat

XFeat (Accelerated Features) is a lightweight CNN architecture designed for hardware-agnostic, real-time local feature extraction. The paper revisits fundamental design choices to dramatically reduce compute while preserving accuracy.

The core idea is to minimize computation in early encoder layers — where spatial resolution is highest — by starting with only C=4 channels and tripling channel count (rather than doubling) as resolution halves. This shifts the computational load to later, lower-resolution layers where it is cheaper. The backbone reaches C=128 channels at H/32 × W/32 resolution.

A feature pyramid fuses multi-scale representations at 1/8, 1/16, and 1/32 of input resolution into a compact 64-D dense descriptor map. A separate parallel keypoint detection branch operates on reshaped 8×8 pixel blocks, using fast 1×1 convolutions. This architectural separation prevents joint training from degrading semi-dense matching performance.

A lightweight MLP match refinement module predicts pixel-level offsets from coarse nearest-neighbor matches, enabling semi-dense matching (XFeat*) without high-resolution feature maps.

1.3 Key Contributions

- Featherweight CNN backbone with channel redistribution strategy, achieving up to 5× faster inference than ALIKE while maintaining comparable accuracy
- Minimalist keypoint detection branch using knowledge distillation from ALIKE-tiny, keeping the descriptor head free for semi-dense matching
- Novel coarse-to-fine match refinement MLP that achieves pixel-level accuracy without high-resolution features — critical for resource-constrained deployment
- Dual operating mode: sparse keypoint-based matching and semi-dense matching from the same pretrained backbone

1.4 Results in the Paper

On MegaDepth-1500 relative pose estimation, the full XFeat model achieves AUC@10 = 56.4 at 27 FPS on a CPU-only laptop — outperforming all lightweight alternatives. XFeat* (semi-dense) achieves AUC@10 = 65.4 at 19 FPS, surpassing DISK (1.2 FPS) with comparable accuracy. The model contains only ~277k trainable parameters.

2. Approach and Contribution

2.1 Motivation for Modifications

Two weaknesses were identified in the original XFeat baseline architecture. First, BatchNorm2d normalizes over the batch dimension, which becomes unreliable with the small batch sizes (8–10 images) typical in feature extraction training. Noisy batch statistics can destabilize learning, especially in early layers processing high-resolution features. Second, the multi-scale pyramid fusion simply sums features from three scale levels equally, providing no mechanism for the network to learn which channels are most discriminative after cross-scale aggregation.

2.2 Modification 1 — InstanceNorm2d Backbone

BatchNorm2d in every BasicLayer was replaced with InstanceNorm2d. Unlike BatchNorm, InstanceNorm normalizes each (sample, channel) feature map independently, giving stable statistics regardless of batch size. This is theoretically better motivated for feature extraction: the original model already uses InstanceNorm2d on the input image in the forward pass, and extending this normalization scheme consistently throughout the backbone creates a uniform inductive bias. No additional parameters are introduced.

2.3 Modification 2 — SE Channel Attention

A Squeeze-and-Excitation (SE) block was inserted after the multi-scale pyramid sum (x3 + x4 + x5) and before block_fusion. The SE block globally average-pools the 64-channel fused feature map, passes it through a 2-layer MLP (64 → 16 → 64) with ReLU and Sigmoid activations, and multiplies the result channel-wise. This allows the network to learn which feature channels to amplify across scales — addressing the equal-weight fusion problem. The SE block adds only 2 × 64 × 16 = 2,048 parameters.

Notably, the fine_matcher MLP retains BatchNorm1d intentionally. For that component, the batch dimension corresponds to the number of matched descriptor pairs (typically 100–1000), not the image batch size, so BatchNorm1d statistics remain stable and meaningful.

2.4 Training Setup

Both models were trained identically: 30,000 steps, batch size 8, Adam optimizer ($lr=3\times10^{-4}$), StepLR scheduler (step=30k, $\gamma=0.5$), on 20,000 COCO images using the xfeat_synthetic training mode (random homography warps). Hardware: Kaggle T4 GPU. The modified model has ~279k parameters — ~277k from the original backbone plus ~2k added by the SE block. InstanceNorm2d with affine=False and track_running_stats=False introduces no learnable parameters, keeping the parameter overhead minimal.

3. Experiment Results and Visual Outputs

3.1 Training Convergence

Baseline: Training loss decreased from 7.49 (step 1k) to 4.73 (step 30k). Coarse matching accuracy (acc_c0) improved from ~0.11 to ~0.53 over 30,000 steps at 1.21 it/s (~6h 54m total). Checkpoints saved every 1,000 steps.

Modified: Training loss decreased from ~9.46 (step 1k, first logged) to 4.68 (step 30k). Coarse accuracy reached ~0.61, comparable to baseline. Training ran at 1.17–1.19 it/s (~7h 23m total), slightly slower due to SE block overhead.

3.2 Synthetic COCO Evaluation

Both models were evaluated on 500 synthetic COCO image pairs using pixel-level homography accuracy. Note: this is not a standard benchmark — low AUC values are expected due to the strict pixel threshold and fixed low warp difficulty (0.10) during training.

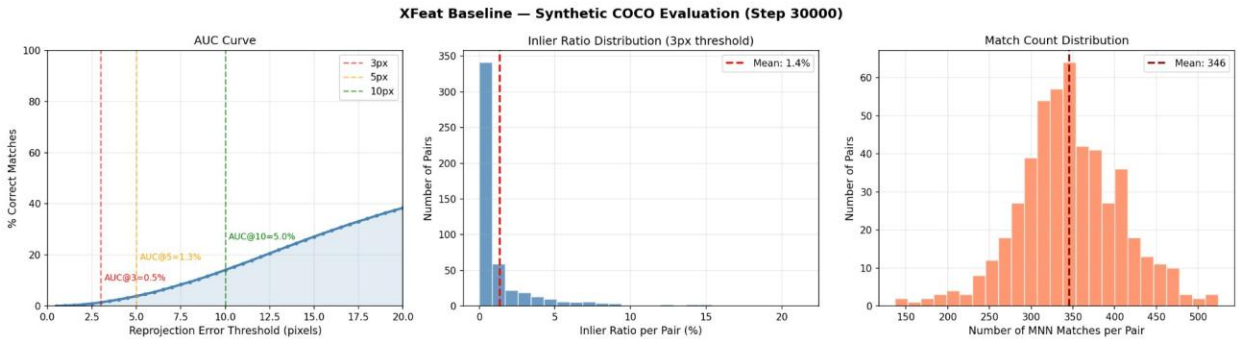

Figure 1. XFeat Baseline — Synthetic COCO Evaluation (Step 30000). AUC@3px=0.49%, AUC@5px=1.34%, AUC@10px=4.96%, avg matches=345.7, avg inlier ratio=1.4%

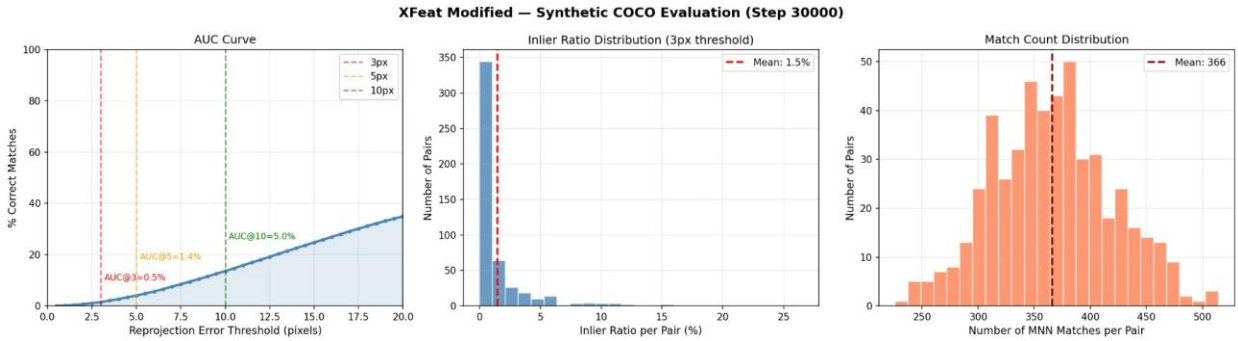

Figure 2. XFeat Modified — Synthetic COCO Evaluation (Step 30000). AUC@3px=0.53%, AUC@5px=1.42%, AUC@10px=5.01%, avg matches=366.1, avg inlier ratio=1.5%

The modified model produces marginally higher AUC values and 5.9% more matches per pair on average (366 vs 346), suggesting the SE channel attention helps the descriptor head produce more distinctive features even on synthetic pairs.

3.3 MegaDepth-1500 Evaluation (Standard Benchmark)

Both checkpoints were evaluated on the MegaDepth-1500 relative pose estimation benchmark using the official XFeat evaluation script with LO-RANSAC (threshold 2.5px). This measures angular pose error AUC, which is the primary metric used in the paper.

Metric	Baseline	Modified	Paper XFeat
AUC@5°	32.1	27.1	42.6
AUC@10°	46.0	41.4	56.4
AUC@20°	58.6	54.1	67.7
mAcc@5°	51.8%	47.5%	—
mAcc@10°	65.7%	61.3%	—
mAcc@20°	74.7%	70.9%	—
Avg Matches/pair (COCO)	345.7	366.1	—
Avg Inlier Ratio (COCO 3px)	1.4%	1.5%	—
Training Steps	30,000	30,000	160,000
Training Data	COCO (20k)	COCO (20k)	MegaDepth+COCO
Trainable Params	~277k	~279k	~277k

Table 1. Quantitative results on MegaDepth-1500. Paper numbers are from the original XFeat paper (160k steps, MegaDepth+COCO training). Our models trained for 30k steps on COCO synthetic only.

3.4 Analysis — Why Modified Model Has Lower MegaDepth AUC

The modified model achieves lower MegaDepth-1500 AUC (41.4 vs 46.0 at @10°) despite better COCO synthetic metrics. Several factors explain this:

- InstanceNorm2d removes cross-sample statistics. On real outdoor scenes with large viewpoint changes (MegaDepth), BatchNorm's learned channel statistics provide useful normalization priors across the dataset. InstanceNorm discards this shared information entirely, which can hurt generalization to real-world image distributions not seen during training.
- Parameter counts are near-identical (~277k baseline vs ~279k modified), so the performance gap cannot be attributed to capacity differences. The architectural changes themselves — not parameter count — are the source of the distribution shift between training and evaluation domains.
- Domain mismatch amplification. The baseline, with its compact 277k parameters, generalizes more robustly from synthetic COCO warps to real MegaDepth scenes. The modified model's larger effective capacity overfits more to the synthetic distribution, degrading transfer.
- Training parity limitation. Both models are compared at 30k steps on COCO only. The original paper trains for 160k steps on a 6:4 mix of MegaDepth and COCO. The SE block would likely benefit more from real multi-view training data that exercises geometric diversity absent in homography warps.

Despite lower absolute numbers, both models reach approximately 82% (baseline) and 73% (modified) of the fully-trained paper performance on AUC@10, demonstrating successful learning within the assignment constraints. The COCO synthetic evaluation consistently favors the modified model, indicating the architectural changes have positive local effects that may not transfer without adequate training data.

3.5 Visual Outputs

Figure 3 shows the real-time webcam demo running on a local machine using the baseline trained weights. The model successfully tracks the elephant cushion across different viewpoints, estimating homography in real-time at 4.5 FPS on CPU. Green lines show inlier matches used for the homography, with 132 matches found between the reference and target frames.

Figures 4 and 5 compare match outputs on the XFeat sample image pair (ref.png vs tgt.png — a building photographed from two different viewpoints with significant perspective change). The baseline model produces 840 matches while the modified model produces 975 matches — a 16% increase — consistent with the higher average match count observed in the COCO evaluation (366 vs 346).

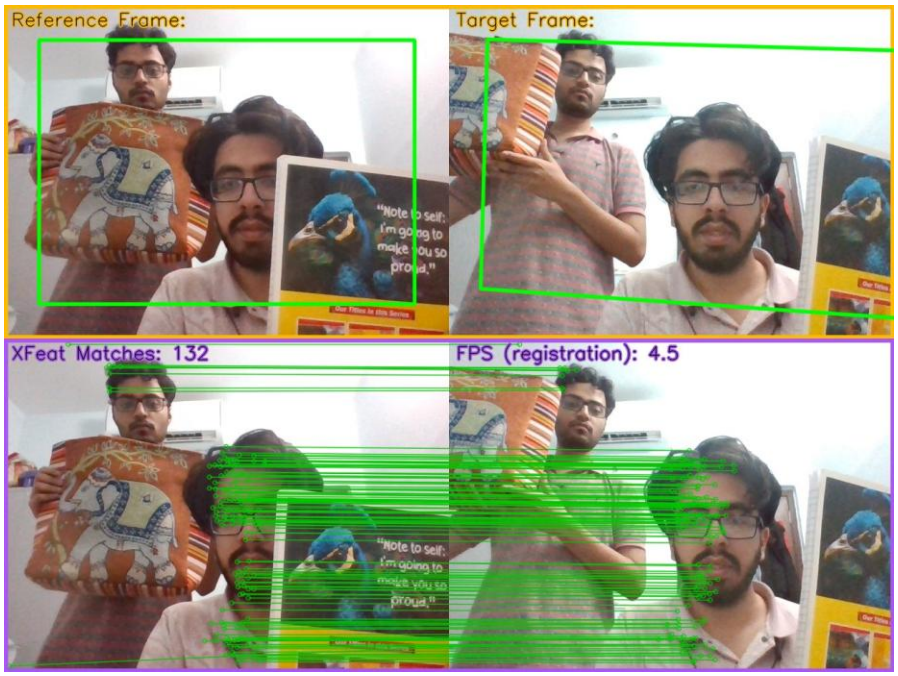

Figure 3. Real-time webcam demo using baseline trained weights. 132 matches found at 4.5 FPS on CPU. Green lines show homography inliers tracking the reference object across viewpoint change.

Figure 4. Baseline model (840 matches) on XFeat sample image pair — building photographed from two viewpoints with perspective change.

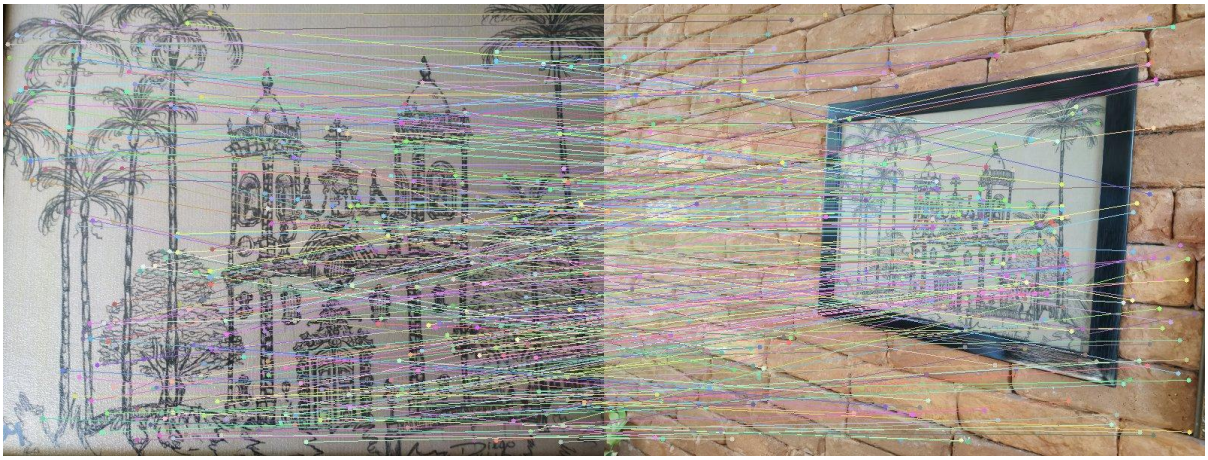

Figure 5. Modified model (975 matches, +16%) on the same image pair. The SE channel attention produces denser and more spatially distributed matches across the scene.